Exploring County-Level Socioeconomic Factors Behind Voter Registration in the U.S. Rust Belt

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Table of Contents

1 Abstract

This project investigates the extent to which county-level socioeconomic characteristics predict estimated voter registration rates across counties in the U.S. Rust Belt in 2020. Using datasets from the U.S. Census Bureau, we construct three regression models to relate voter registration rates to factors capturing educational composition, economic conditions, and demographic characteristics. We find that measures of education and unemployment exhibit the strongest and most consistent associations with registration, as both educational attainment and unemployment rates are significant predictors in estimating voter registration rates. Overall, our results show that socioeconomic composition meaningfully explains variation in voter registration rates, while leaving room for additional political and institutional factors that are not captured in our models.

2 Introduction

The COVID-19 pandemic represents an unprecedented period within the United States and the broader global economy. Issues ranging from supply chain shocks, economic shutdowns, and vaccination distribution further ignited an already polarized nation heading into another presidential election. This changed the way people were able to participate in civic engagement, making this election consequential beyond the need to survive the pandemic. Compared to previous elections, the concern of voter registration and representation became a significantly larger topic, as campaigns across social media further emphasized the call for high civic engagement.

Particularly, U.S. Rust Belt represents a core sample of highly sought voters across all presidential candidates. This region of the United States represents an economically diverse and politically competitive range, stemming from its significant mix of deindustrialized counties containing different conditions of economic recovery. Our research seeks to study the extent to which county-level socioeconomic characteristics predict estimated voter registration rates across counties within the U.S. Rust Belt. This study not only captures relationships regarding political representation and inequality, but also identifies socioeconomic factors that either encourage or prevent civic engagement.

Our contribution provides a county-level test of whether socioeconomic conditions predict estimated voter registration rates across counties within the U.S. Rust Belt during the 2020 election. Using a cross-sectional regression framework, we relate regression rates to broad categories of predictors, such as educational attainment, economic conditions, and demographic composition. We use these categories to evaluate which types of county characteristics are most consistently associated with higher or lower-level voter registration. This approach moves beyond statewide averages by explaining variation across counties, offering a clearer picture of how local inequality and opportunity may shape civic engagement during an unusually disruptive election environment.

3 Data

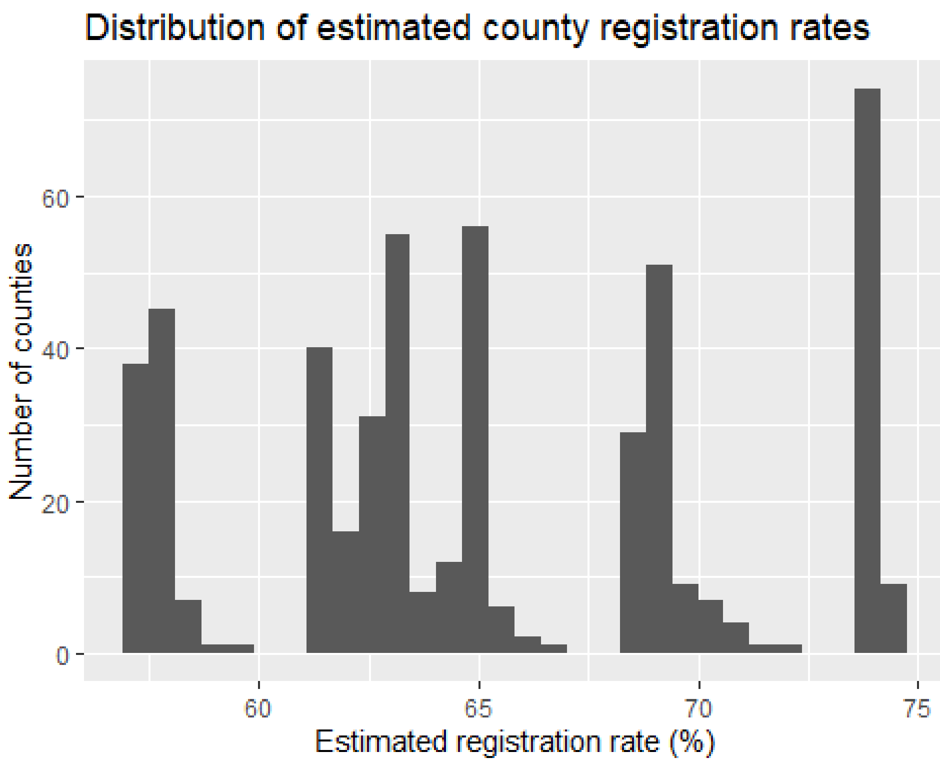
This study defines the U.S. Rust Belt by the following states: Indiana, Illinois, Michigan, Ohio, Pennsylvania, and Wisconsin. In total, our dataset contains 504 counties, providing an adequate sample of the U.S. Rust Belt that further reflects common characteristics among the citizen voting age population of the United States. The data for our project is collected solely from the U.S. Census Bureau, using American Community Survey (ACS) 5-year estimates to ensure consistent county-level coverage and more reliable estimates for smaller populations.

The response variable for this study required the construction of a proxy variable that measures county-level voter registration rates. This is constructed using the CVAP 2019-2023 and Voting and Registration datasets. This variable measures county-level estimated voter registration rates among the citizen voting-age population in 2020.

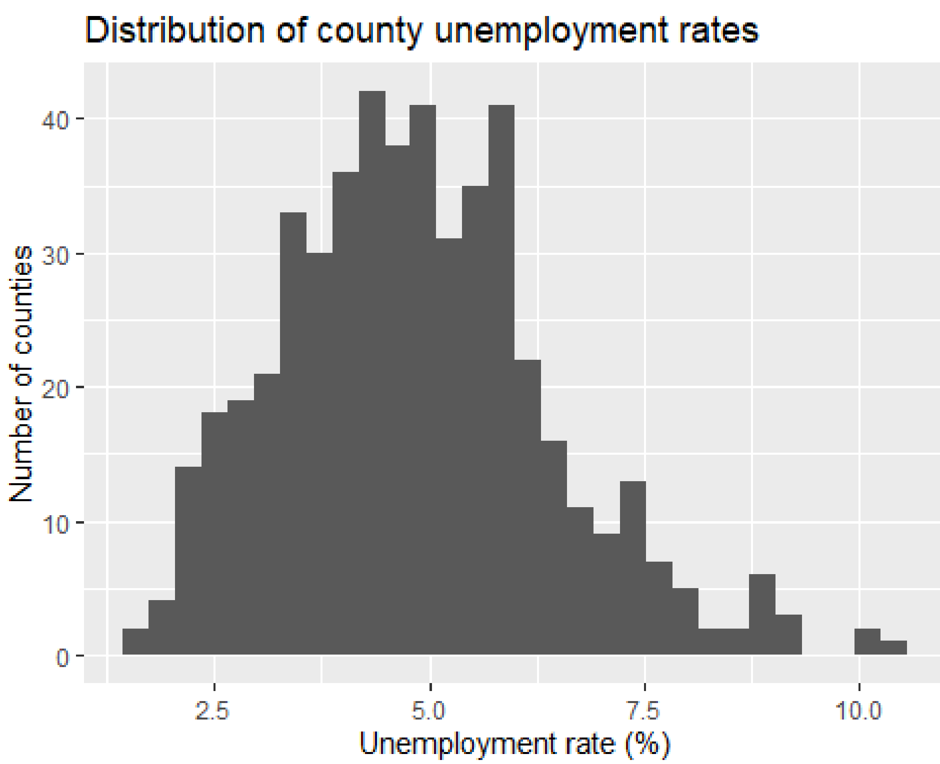
This study incorporates several explanatory variables, gathered from the following datasets: Educational attainment (S1501), Poverty (S1701), Income (S1901), Employment status (S2301), and Demographic and housing estimates (DP05). From these sources, we construct predictors capturing educational attainment (e.g., populations who have completed different levels of education), local economic conditions (e.g., unemployment and poverty rates), and demographic structure (e.g., age, gender, and ethnicity). All predictors are measured at the county level and are reported in percentage points unless otherwise specified.

Univariate summaries for key variables (county level)

variable	n	mean	sd	min	p25	median	p75	max
Percent_9_12_NoDiploma	504	6.700	2.138	1.200	5.275	6.60	7.900	15.300
Percent_College_NoDegree	504	20.709	3.351	9.800	18.600	21.10	23.225	29.100
Percent_HS_Diploma	504	37.087	6.955	14.700	32.500	37.65	42.100	54.300
Percent_Poverty	504	12.367	4.427	2.600	9.300	11.90	15.000	33.100
Unemployment_Rate	504	4.831	1.583	1.600	3.700	4.70	5.800	10.400
edu_ba_plus_pct	504	22.372	8.616	9.400	16.400	19.70	25.800	59.400
pct_65plus	504	19.516	3.799	10.400	17.400	19.00	21.025	36.300
pct_hispanic	504	4.118	4.070	0.100	1.700	2.70	4.900	32.000
pct_male	504	50.085	1.796	47.100	49.200	49.70	50.400	69.500
reg_rate_all_pct	504	65.199	5.370	57.212	61.447	64.63	69.002	74.445



Distribution of estimated county registration rates
Number of counties
0
20
40
60
60
65
70
75
Estimated registration rate (%)



Distribution of county unemployment rates
Number of counties
0
10
20
30
40
2.5
5.0
7.5
10.0
Unemployment rate (%)

3.0.1 Univariate Analysis

The tables above summarize the county-level distribution of the dependent variable and key predictors across the 504 Rust Belt counties in our sample. Estimated county voter registration rates average 65.2%, with most counties falling roughly between 61.4% and 69.0%. The histogram shows noticeable clustering across the range of registration rates, which is a side effect of the construction of our proxy variable. This will be expanded upon through our models.

Among predictors, unemployment rates are centered around 4.8%, with most counties between 3.7% and 5.8%, but show a right tail extending above 10%. Poverty averages 12.4%, and educational composition varies substantially across counties (e.g., high school diploma shares average 37.1%, while bachelor's degree or higher averages 22.4%). Overall, these univariate patterns indicate meaningful cross-county variation in both registration and socioeconomic characteristics, supporting the use of multivariate regression to assess which factors are most strongly associated with voter registration.

4 Hypotheses

We test whether county-level socioeconomic characteristics are associated with estimated voter registration rates.

H_0 :No systematic relationship between registration and socioeconomic characteristics.
H_A :At least one socioeconomic characteristic is significantly associated with registration.
H_1 :Education and income-related measures have the strongest association with registration.

Consistent with prior work on political participation, we expect education and income-related measures to exhibit the strongest associations with registration.

5 Analysis

This project will report three linear regression models. For our model selection, we will be using backwards elimination, represented by our baseline model. In Model 0, we fed all 23 of our independent variables into the model.

5.1 Model 0

$$\text{reg\_rate\_all\_pct}_i = \beta_0 + \sum_{k=1}^{23} \beta_k X_{ki} + \varepsilon_i$$

Model 0 Regression Results

	(1)
(Intercept)	66.327
	(221.071)
edu_hs_plus_pct	-0.726
	(2.209)
edu_somecoll_pct	0.369***
	(0.106)
edu_ba_plus_pct	0.141
	(0.100)
edu_grad_plus_pct	0.096
	(0.193)
edu_lt_hs_pct	-1.125
	(2.213)
Percent_Less_Than_9th	0.753***
	(0.179)
Percent_College_NoDegree	0.353**
	(0.131)
Percent_Poverty	0.133
	(0.087)
Unemployment_Rate	0.699***

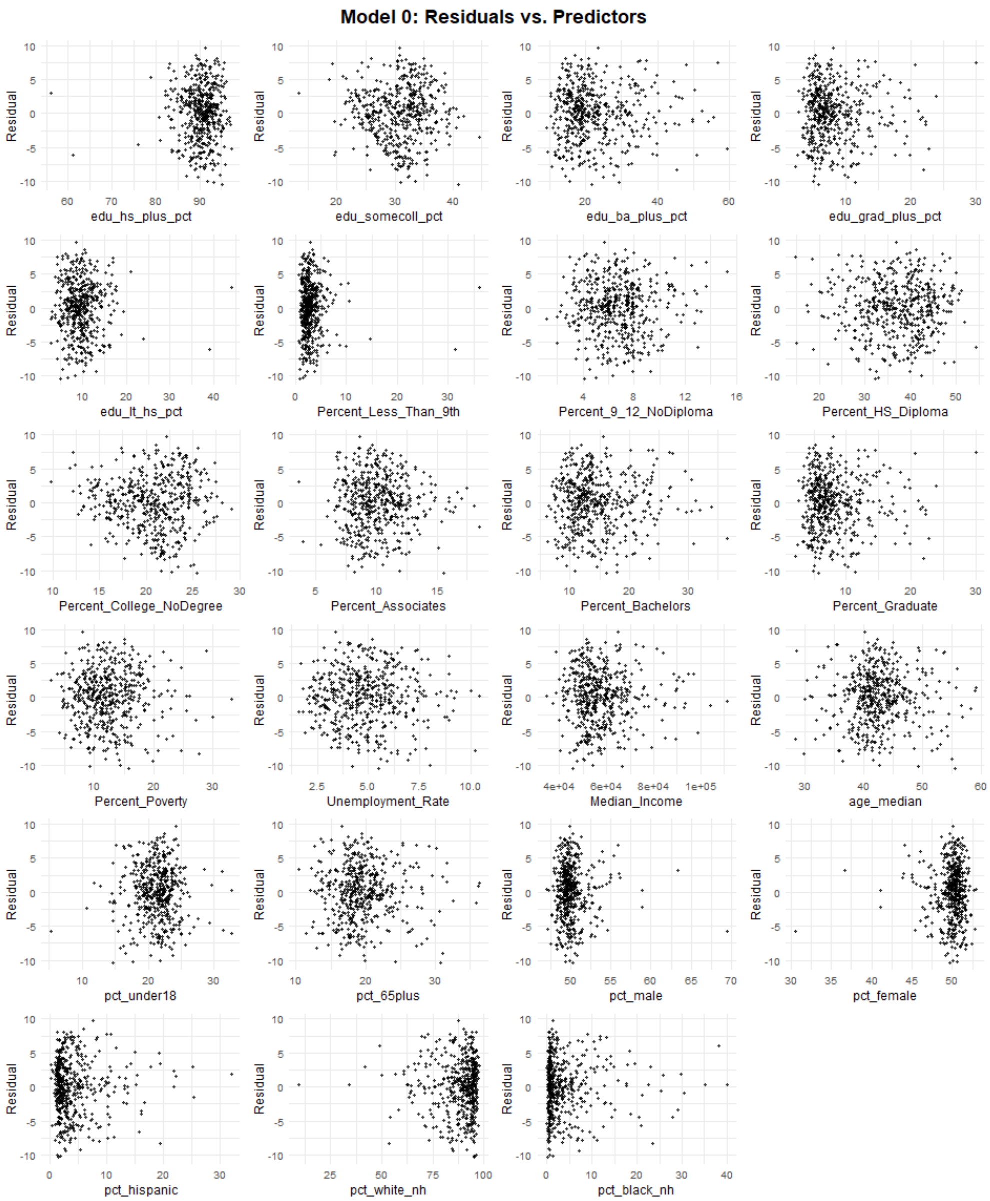
	(1)
	(0.164)
Median_Income	0.000
	(0.000)
age_median	0.050
	(0.169)
pct_under18	0.094
	(0.132)
pct_65plus	0.329
	(0.212)
pct_male	0.587***
	(0.133)
pct_hispanic	0.210**
	(0.071)
pct_white_nh	0.047
	(0.046)
pct_black_nh	0.027
	(0.061)
Num.Obs.	504
R2	0.471
R2 Adj.	0.452
AIC	2841.0
BIC	2921.2
Log.Lik.	-1401.494
RMSE	3.90

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

The summary shows Model 0 is not the best fit for predicting voter registration rate as only 6 of our 23 predictors showed significance. However, the adjusted r-squared value is strong, represented by a value of 0.45.

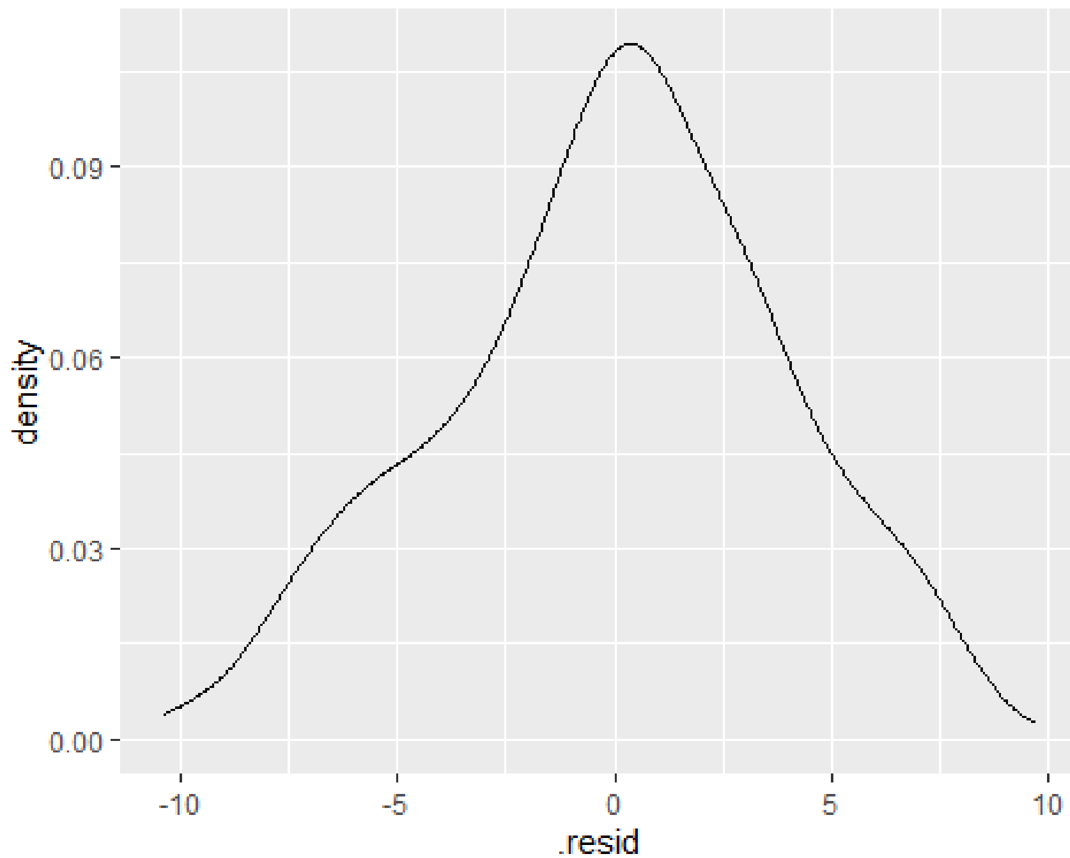
5.1.1 Conditions for Model 0

To check for the condition of linearity, we plot all of our 23 independent variables against the residuals, represented by the following figure:



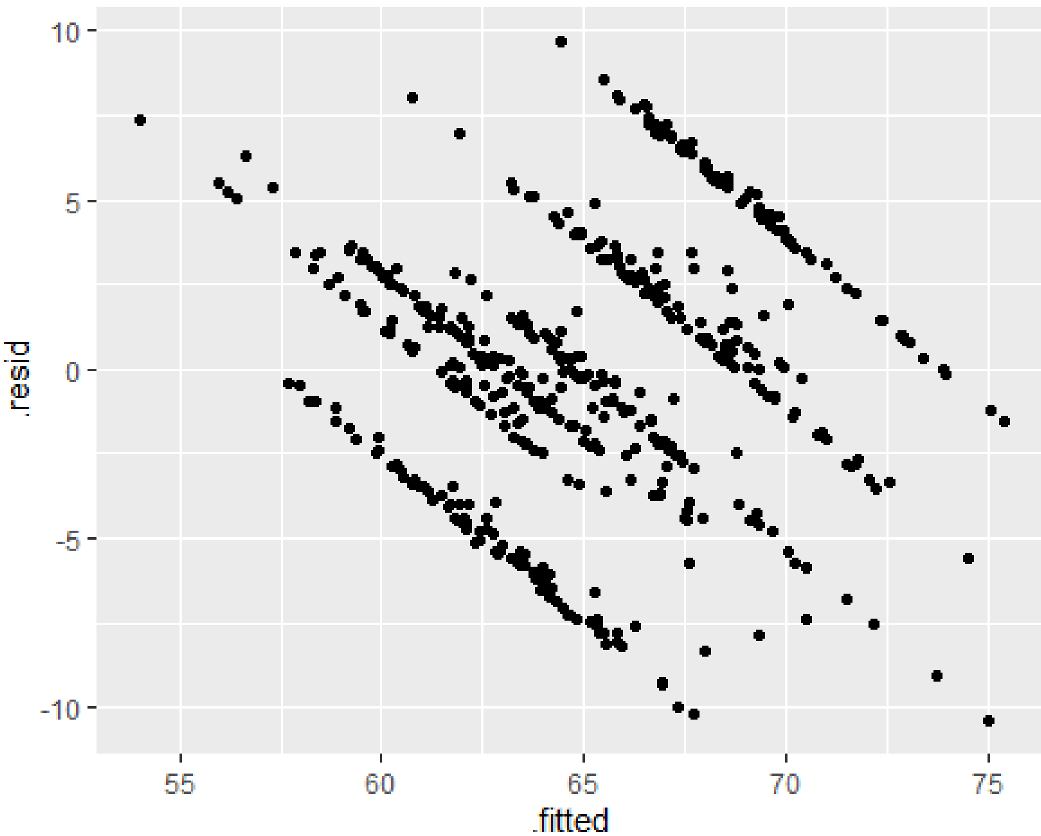
The figures show that most of the patterns look randomly scattered. A couple of the plots reveal some clustering towards one side of the graph, due to the presence of outliers. This does not give any cause for concern. However, to refine the model, one could alter the presence of outliers by removing them to make the patterns more random than they appear in our current sample.

To check for the condition of nearly normal residuals, we plot the distribution of the residuals, represented by the following figure:



The density plot of Model 0's residuals appears normal, satisfying the condition of nearly normal residuals.

To check for the condition of equal variability, we plot our fitted values against our residuals, looking for more random patterns similar to what was previously displayed in the linear plots:



This plot gives an unexpected output, revealing a striped pattern that is not a randomized pattern. We see this occur in all of our equal variance plots, which is elaborated further by our final model. With this, we discuss whether the model will satisfy the condition of equal variance.

5.2 Model 1

Our next model builds upon backwards elimination, seeking to improve the significance of our predictors. Model 1 is a linear model that reduces the 23 predictors to 6, focusing on the significance of core variables.

$$\text{reg\_rate\_all\_pct}_i = \beta_0 + \sum_{k=1}^{6} \beta_k X_{ki} + \varepsilon_i$$

Model 1 Regression Results

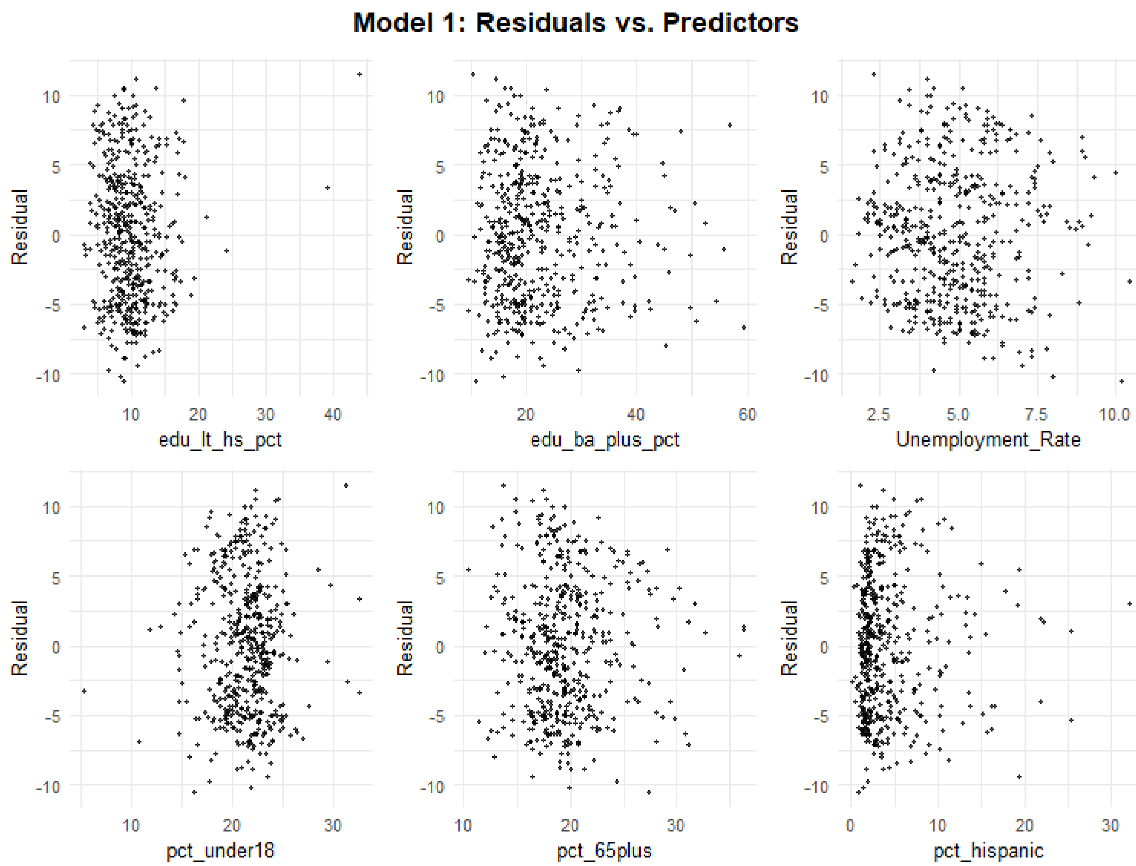
	(1)
(Intercept)	52.131***
	(4.158)
edu_lt_hs_pct	-0.231**
	(0.077)
edu_ba_plus_pct	0.064+
	(0.036)
Unemployment_Rate	0.898***
	(0.144)
pct_under18	0.016
	(0.111)
pct_65plus	0.424***
	(0.084)
pct_hispanic	0.224***
	(0.059)
Num.Obs.	504
R2	0.211
R2 Adj.	0.201
AIC	3020.1
BIC	3053.9
Log.Lik.	-1502.039
F	22.151
RMSE	4.77

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The summary shows that Model 1 improves the significance of our predictors. Out of the 6 independent variables, 5 show significance in their ability to predict voter registration. However, the adjusted r-squared of Model 1 is noticeably lower than our baseline model. Between Model 0 and Model 1, the adjusted r-squared fell from 0.45 to 0.20. To improve this, our final model utilizes a log transformation on our response variable.

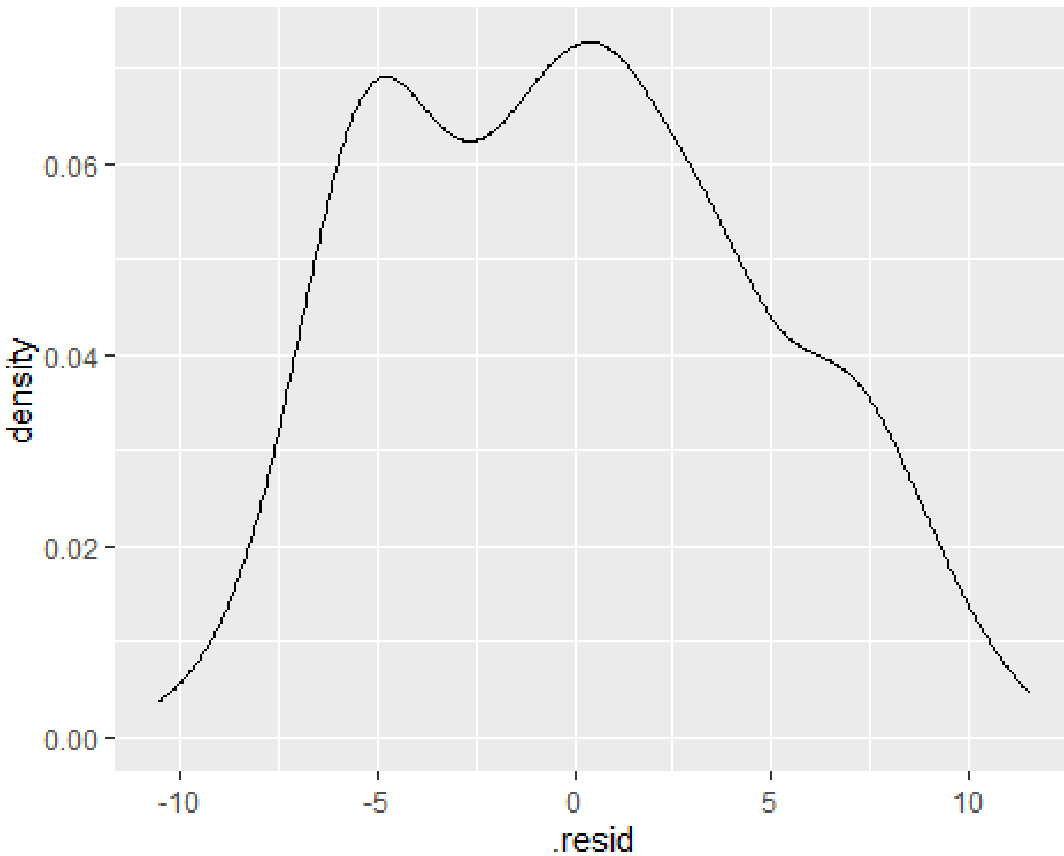
5.2.1 Conditions for Model 1

To check for the condition of linearity, we plot our independent variables against the model's residuals, represented by the following figure:



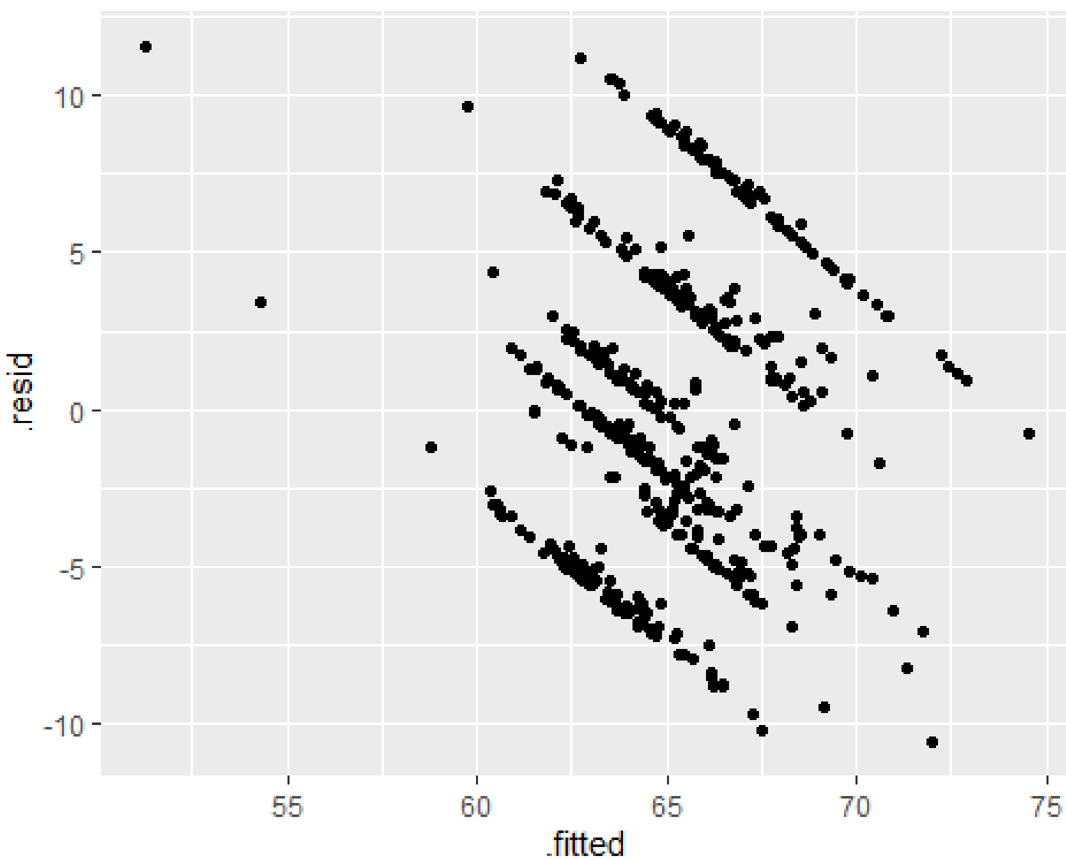
Similar to the baseline model, we see randomly scattered patterns. The condition of linearity is satisfied for Model 1.

To check for the condition of nearly normal residuals, we plot the distribution of the model's residuals, represented by the following figure:



Our residual distribution does not look as normal as it did in the baseline model, but it does show relative symmetry around 0. Because of this, the condition of nearly normal residuals is satisfied in Model 1.

To check for the condition of equal variability, we plot the fitted values of the model against the residuals, represented by the following figure:



Similar to the baseline model, we see the striped pattern in our output. This will be explained in the conditions of our final model. Despite the emergence of this pattern, Model 1 satisfies the condition of equal variance.

5.3 Model 2

The final model represents a log-linear regression that improves the adjusted r-squared value while solidifying significant predictors. Model 2 possesses a log-transformed dependent variable of the voter registration rate, building upon backward elimination from the baseline model.

$$\log(\text{reg\_rate\_all\_pct}_i) = \beta_0 + \sum_{k=1}^{9} \beta_k X_{ki} + \varepsilon_i$$

Model 2 Regression Results (log dependent variable)

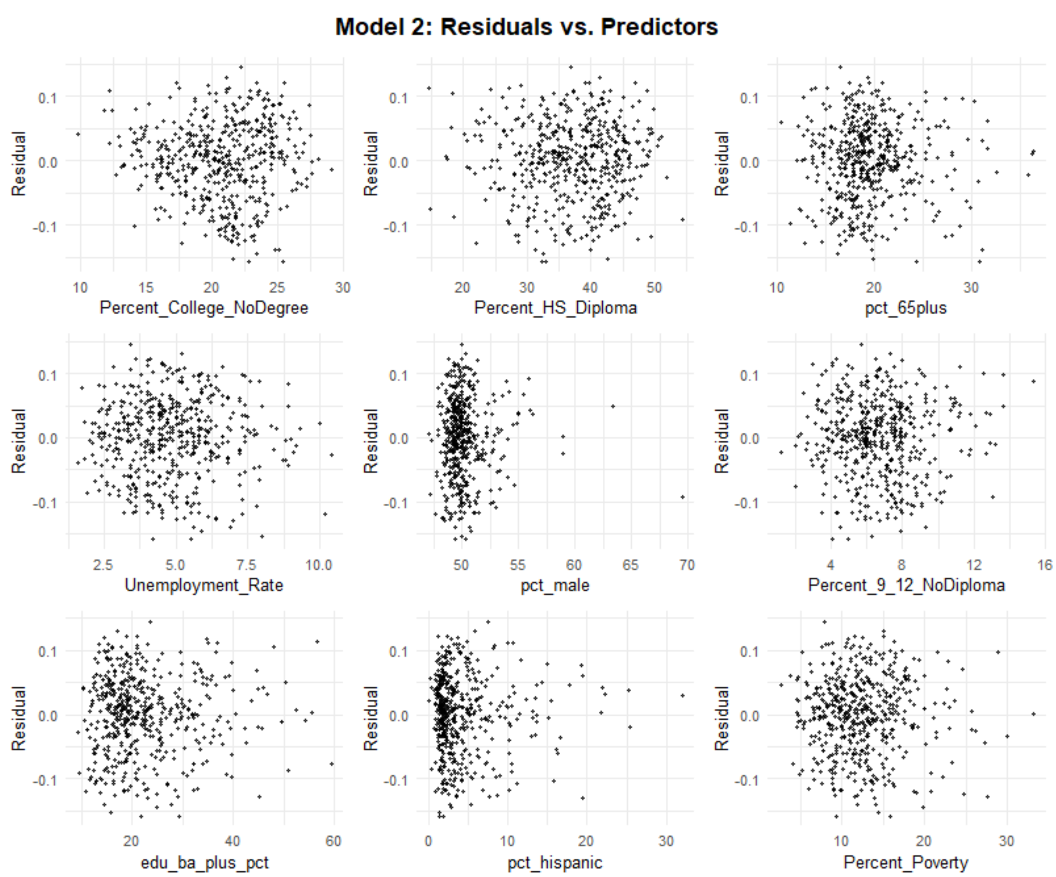
	(1)
(Intercept)	3.848***
	(0.114)
Percent_College_NoDegree	0.005***
	(0.001)
Percent_HS_Diploma	-0.006***
	(0.001)
pct_65plus	0.005***
	(0.001)
Unemployment_Rate	0.010***
	(0.002)
pct_male	0.008***
	(0.002)
Percent_9_12_NoDiploma	-0.012***
	(0.002)
edu_ba_plus_pct	-0.003**
	(0.001)
pct_hispanic	0.002**
	(0.001)
Percent_Poverty	0.001+
	(0.001)
Num.Obs.	504
R2	0.464
R2 Adj.	0.454
AIC	2823.0
BIC	2869.5
Log.Lik.	703.224
F	47.518
RMSE	0.06

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

The summary reveals significant improvements in the construction of Model 2. The adjusted r-squared increased from the previous model, moving from 0.20 to 0.45. Additionally, 8 out of 9 independent variables show significance in their ability to predict the voter registration rate.

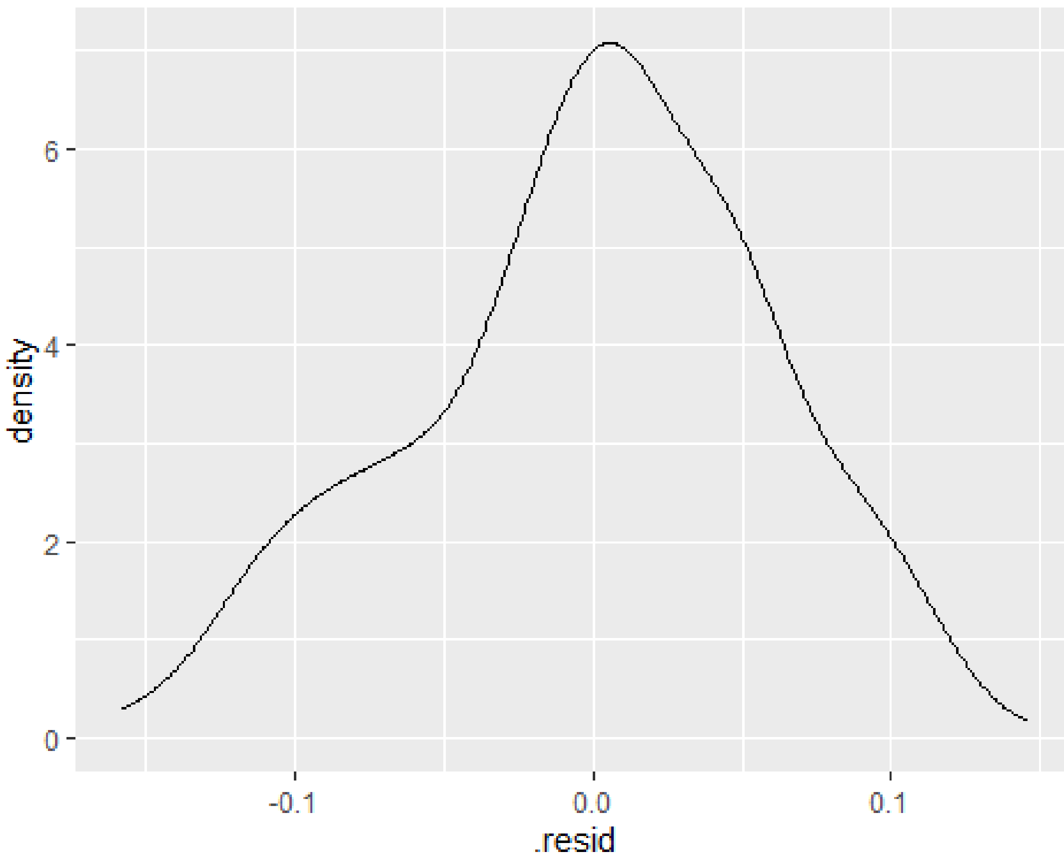
5.3.1 Conditions for Model 2

To check for the condition of linearity, we plot our independent variables against the residuals, represented by the following figure:



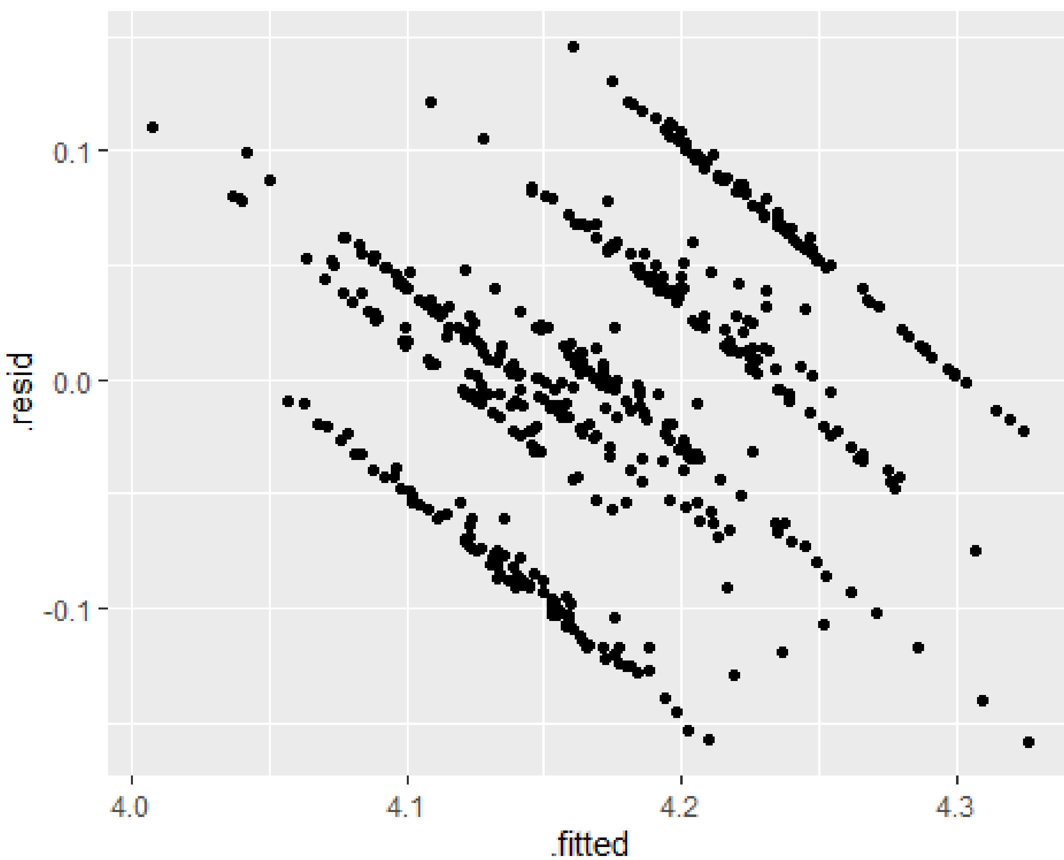
We see similar random patterns resembling the previous models, with only two graphs showing clustering towards one side due to outliers. Despite the existence of these outliers, the condition of linearity is satisfied for Model 2.

To check the condition of nearly normal residuals, we plot the distribution of the residuals, represented by the following figure:



The distribution is symmetric around 0, ensuring the condition of nearly normal residuals is satisfied for Model 2.

Lastly, to check the condition of equal variance, we plot our fitted values against our residuals, represented by the following figure:



Similar to Model 0 and Model 1, we see the pattern emerge.

This pattern emerges across each of the fitted vs residual plots, no matter the independent variables that were selected in each model. This occurs because of how the dependent variable was constructed as a proxy variable. Because our regression model produces continuous fitted values, subtracting these predictions from a dependent variable that is recorded at a fixed decimal causes the residuals to cluster at specific values. Each rounded value of the voter registration rate corresponds to one of the horizontal bands seen in the fitted versus residuals plot. Since the overall vertical spread of the residuals remains relatively constant across the range of fitted values, this pattern does not indicate unequal variance.

Therefore, the condition of equal variance is satisfied for Model 2.

6 Results

Out of the three models we estimated, Model 2 provides the strongest overall fit given the conditions, with an improved adjusted r-squared and smaller p-values relative to the previous models. For these reasons, we select Model 2 as our primary model for interpretation. However, Model 2 includes multiple education variables that are conceptually related, so we check multicollinearity using variance inflation factors (VIFs).

Variance Inflation Factors (VIFs) for Model 2

Predictor	VIF	Collinearity
edu_ba_plus_pct	9.85	Moderate
Percent_HS_Diploma	8.14	Moderate
Percent_College_NoDegree	2.91	Low
Percent_9_12_NoDiploma	2.62	Low
Unemployment_Rate	1.90	Low
Percent_Poverty	1.82	Low
pct_65plus	1.41	Low
pct_hispanic	1.33	Low
pct_male	1.17	Low

The VIF results indicate moderate collinearity, particularly among the education measures (e.g., Percent_HS_Diploma and edu_ba_plus_pct). While these values remain below the common threshold of 10, they are close enough that we interpret individual education coefficients cautiously, since collinearity can inflate standard errors and produce counterintuitive signs when several overlapping measures are included together.

Because the dependent variable is the natural log of the estimated registration rate, Model 2 is a log-linear regression. Coefficients can therefore be interpreted approximately as semi-elasticities: a one-unit increase in a predictor is associated with an expected change of $100 \times \beta\%$ in the registration rate, holding other covariates constant.

In Model 2, we test H_0 that there is no relationship between county-level socioeconomic characteristics and estimated voter registration rates. The regression results provide evidence against this null, as the model explains a meaningful share of cross-county variation, and multiple predictors are statistically distinguishable from zero. We therefore reject H_0 in favor of H_A that at least one socioeconomic characteristic is associated with registration.

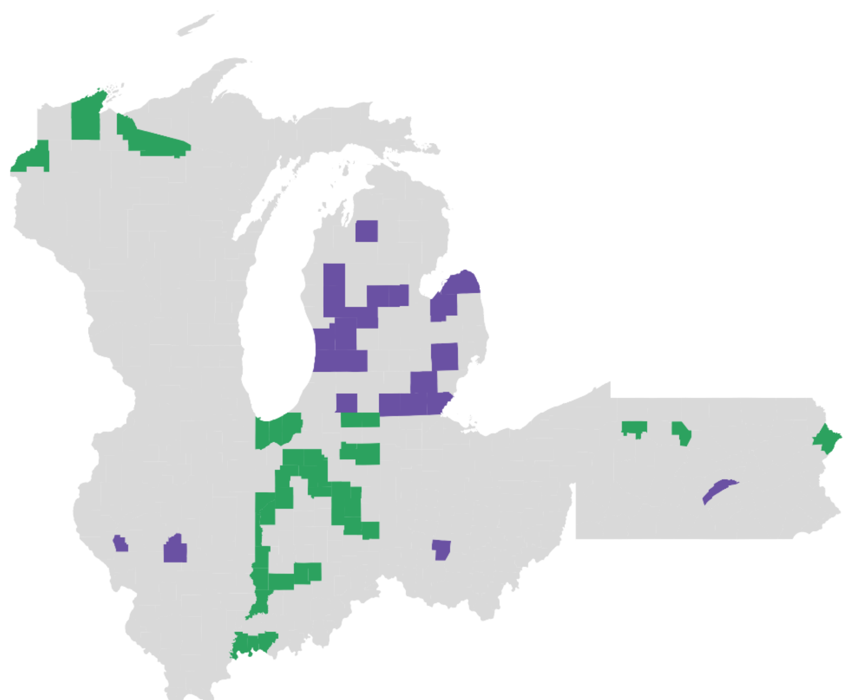
We also evaluate H_1 that education and income-related measures have the strongest association with registration. Consistent with H_1, several education and economic indicators appear among the most substantively meaningful predictors, although interpretation of individual education coefficients should be cautious because multiple, conceptually overlapping education measures are included simultaneously and may share explanatory power.

Substantively, the two strongest associations in Model 2 are Percent_9_12_NoDiploma and the unemployment rate. Counties with larger shares of residents whose highest schooling is grades 9–12 with no diploma tend to have lower estimated registration rates, aligning with the broader expectation that lower educational attainment is associated with weaker civic engagement. In contrast, counties with higher unemployment tend to have higher estimated registration rates in this sample. While this may appear counterintuitive, it is plausible in 2020 given the salience of economic policy during the pandemic and widespread labor-market disruption.

Other covariates in Model 2 generally move in theoretically consistent directions. For example, registration tends to be higher in counties with older populations and varies systematically with educational composition. However, these effects are smaller or less straightforward to interpret when multiple related predictors are included together. Overall, Model 2 indicates that socioeconomic and demographic characteristics are meaningfully associated with estimated county registration rates across Rust Belt counties in 2020, while also leaving room for additional political and institutional factors not captured in the model.

6.0.1 Geographic Distribution of Model Error

Model 2: Counties flagged by standardized residuals (90% level)



Higher than predicted · Lower than predicted · Not significant

Model 2: Counties flagged by standardized residuals (abs(r) > 1.645; 90% level)

State	County	GEOID	Direction	Std. residual
MI	Allegan County	26005	Higher than predicted	2.42
MI	Newaygo County	26123	Higher than predicted	2.15
IL	Scott County	17171	Higher than predicted	2.03
MI	Hillsdale County	26059	Higher than predicted	2.02
MI	Montcalm County	26117	Higher than predicted	2.01
MI	Midland County	26111	Higher than predicted	1.94
MI	Washtenaw County	26161	Higher than predicted	1.93
MI	Ottawa County	26139	Higher than predicted	1.87
MI	Kent County	26081	Higher than predicted	1.85
PA	Juniata County	42067	Higher than predicted	1.85
MI	Tuscola County	26157	Higher than predicted	1.82
MI	Barry County	26015	Higher than predicted	1.82
MI	Lenawee County	26091	Higher than predicted	1.79
MI	Huron County	26063	Higher than predicted	1.77
MI	Oakland County	26125	Higher than predicted	1.75
IL	Christian County	17021	Higher than predicted	1.75
MI	Monroe County	26115	Higher than predicted	1.73
MI	St. Joseph County	26149	Higher than predicted	1.72
OH	Fayette County	39047	Higher than predicted	1.67
MI	Kalkaska County	26079	Higher than predicted	1.67
MI	Isabella County	26073	Higher than predicted	1.67
MI	Lake County	26085	Higher than predicted	1.66
IN	Tippecanoe County	18157	Lower than predicted	-1.65
IN	Miami County	18103	Lower than predicted	-1.66
IN	Fountain County	18045	Lower than predicted	-1.68
WI	Burnett County	55013	Lower than predicted	-1.70
IN	Pulaski County	18131	Lower than predicted	-1.70
IN	Henry County	18065	Lower than predicted	-1.71
IN	Madison County	18095	Lower than predicted	-1.71
IN	Cass County	18017	Lower than predicted	-1.74
PA	Forest County	42053	Lower than predicted	-1.75
IN	Whitley County	18183	Lower than predicted	-1.76
IN	Greene County	18055	Lower than predicted	-1.77
IN	Knox County	18083	Lower than predicted	-1.78
PA	Pike County	42103	Lower than predicted	-1.79
IN	Sullivan County	18153	Lower than predicted	-1.79
IN	Vanderburgh County	18163	Lower than predicted	-1.81
IN	LaGrange County	18087	Lower than predicted	-1.83

State	County	GEOID	Direction	Std. residual
IN	Wayne County	18177	Lower than predicted	-1.86
IN	Posey County	18129	Lower than predicted	-1.89
IN	Warrick County	18173	Lower than predicted	-1.93
IN	Porter County	18127	Lower than predicted	-1.94
IN	LaPorte County	18091	Lower than predicted	-1.94
IN	Steuben County	18151	Lower than predicted	-1.94
IN	Warren County	18171	Lower than predicted	-1.95
WI	Bayfield County	55007	Lower than predicted	-1.95
IN	Grant County	18053	Lower than predicted	-1.99
PA	Cameron County	42023	Lower than predicted	-2.00
IN	Blackford County	18009	Lower than predicted	-2.04
IN	Howard County	18067	Lower than predicted	-2.07
IN	White County	18181	Lower than predicted	-2.08
IN	Delaware County	18035	Lower than predicted	-2.08
IN	Allen County	18003	Lower than predicted	-2.11
IN	Lake County	18089	Lower than predicted	-2.18
IN	Monroe County	18105	Lower than predicted	-2.18
IN	Fulton County	18049	Lower than predicted	-2.31
WI	Vilas County	55125	Lower than predicted	-2.34
IN	Vigo County	18167	Lower than predicted	-2.41
IN	Vermillion County	18165	Lower than predicted	-2.55
IN	Brown County	18013	Lower than predicted	-2.62
WI	Iron County	55051	Lower than predicted	-2.65

To evaluate whether Model 2 leaves systematic geographic error, Figure X maps county-level standardized residuals from the model and flags counties using a 90% threshold ($|r| > 1.645$). Counties classified as "higher than predicted" are those where observed registration rates exceed the model's fitted values (positive standardized residuals), while "lower than predicted" counties are those where observed registration falls below fitted values (negative standardized residuals). Most counties are not flagged, indicating that the Model 2 covariates capture a substantial share of the broad regional pattern in registration. However, it is important to note that the remaining errors are not uniformly random.

Put simply, the highlighted counties are places where registration is noticeably higher or lower than the model would expect based on the variables included, suggesting that additional local political, administrative, or community-level factors may be influencing registration in those areas.

The map suggests that flagged counties exhibit some geographic clustering, rather than appearing evenly dispersed across the Rust Belt. This clustering is consistent with the possibility that location-specific influences, beyond the socioeconomic and demographic controls included in Model 2, may contribute to voter registration outcomes. These influences could include differences in local election administration, county-level civic infrastructure, institutional rules or practices not measured in our dataset, or remaining measurement error in the registration-rate estimates. The table above provides the corresponding list of flagged counties and the magnitude of their standardized residuals, which helps identify where the model's predictions are most substantially above or below observed rates.

Importantly, this visualization should be interpreted as a diagnostic tool rather than causal evidence. These figures do not identify why a given county deviates from prediction, but instead highlights where Model 2's explanatory variables appear insufficient to fully account for observed registration levels. In that sense, the residual map helps pinpoint potential directions for future work, such as incorporating political and institutional covariates, or modeling spatial dependence explicitly.

7 Conclusion

	Model 0: Full	Model 1: Reduced Model	Model 2: Final Model
College, No Degree (%)	0.353**		0.005***
	SE = 0.131		SE = 0.001
	p = 0.007		p = <0.001
HS Diploma (%)			-0.006***
			SE = 0.001
			p = <0.001
Age 65+ (%)	0.329	0.424***	0.005***
	SE = 0.212	SE = 0.084	SE = 0.001
	p = 0.120	p = <0.001	p = <0.001
Unemployment Rate (%)	0.699***	0.898***	0.010***
	SE = 0.164	SE = 0.144	SE = 0.002
	p = <0.001	p = <0.001	p = <0.001
Male (%)	0.587***		0.008***
	SE = 0.133		SE = 0.002
	p = <0.001		p = <0.001
9–12, No Diploma (%)			-0.012***
			SE = 0.002
			p = <0.001
BA or Higher (%)	0.141	0.064	-0.003**
	SE = 0.100	SE = 0.036	SE = 0.001
	p = 0.160	p = 0.071	p = 0.001
Hispanic (%)	0.210**	0.224***	0.002**
	SE = 0.071	SE = 0.059	SE = 0.001
	p = 0.003	p = <0.001	p = 0.002
Poverty (%)	0.133		0.001
	SE = 0.087		SE = 0.001
	p = 0.125		p = 0.075
Constant	66.327	52.131***	3.848***
	SE = 221.071	SE = 4.158	SE = 0.114
	p = 0.764	p = <0.001	p = <0.001
N	504	504	504
R2	0.471	0.211	0.464
Adj R2	0.452	0.201	0.454

This study asked whether county-level socioeconomic characteristics help predict estimated voter registration rates across the U.S. Rust Belt in 2020. Using U.S. Census–based measures and multiple regression models, we find that socioeconomic factors explain a meaningful share of cross-county variation in estimated registration. Across models, measures related to educational composition and local economic conditions (especially unemployment) show the strongest and most consistent statistical associations with registration.

Among education measures, counties with larger shares of residents with lower educational attainment tend to exhibit lower estimated registration, holding other factors constant. We also find a positive association between county unemployment and estimated registration in 2020, which may reflect the unusual economic and political context of the pandemic. Overall, these results suggest that differences in county socioeconomic composition are strongly related to differences in civic engagement as measured by registration.

Applying a log transformation to the dependent variable improves overall model fit and yields more stable inference in our primary model. Regression diagnostics indicate that the core assumptions for linear regression are reasonably satisfied, as residual plots suggest approximate linearity and near-normal residual behavior. We note that some visual "striping" in residual plots likely reflects rounding in the constructed dependent variable rather than clear evidence of heteroskedasticity.

7.0.1 Limitations

These results should be interpreted with caution. First, the analysis is correlational and does not identify causal effects. We find evidence of associations between socioeconomic conditions and estimated registration, but we cannot conclude that changing any one county characteristic would increase or decrease voter registration.

Second, several predictors exhibit multicollinearity, which can make individual coefficients less stable and sometimes yield counterintuitive signs when overlapping variables are included simultaneously. As a result, we emphasize broader patterns across predictors rather than treating any single education coefficient as definitive.

Third, the dependent variable is a constructed proxy for county registration. Because it is derived from multiple Census-based sources rather than directly observed registration counts, it may contain measurement error and should be understood as an estimate of registration among the citizen voting-age population rather than an official survey.

Finally, the study is limited to one election year (2020) and one region (the Rust Belt). The pandemic, election context, and regional political dynamics may make 2020 unusual, so results should not be assumed to generalize to other years or regions without additional evidence.

7.0.2 Future work

Future research could improve this analysis in several ways. One approach would be to extend the dataset across multiple election years to test whether these relationships persist over time. Another method could incorporate political and institutional variables, such as election administration practices, competitiveness, campaign intensity, or voting access policies in order to better separate socioeconomic correlations from explicitly political drivers of registration. Finally, additional robustness checks (e.g., alternative specifications, sensitivity to outliers, and refined measurement of registration) could strengthen confidence in the findings.

8 References

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Replication code and the constructed dataset are available from the authors upon request.